

Name: _____

PLACE VALUE AND ORDERING INTEGERS, (ROUNDING TO SIGNIFICANT FIGURES)

Date: _____

LO: I can round numbers to a given number of significant figures.

Significant figures

Significant figures (s.f.) are counted from the first non-zero digit. Leading zeros are not significant but trailing zeros after a decimal point are.

Rule: Find the digit at the required position, look at the next digit. If it is 5 or more, round up.

Quick examples

●	3462 to 2 s.f. = 3500	round up (6 5)
●	0.00587 to 2 s.f. = 0.0059	leading zeros not significant
●	45.83 to 1 s.f. = 50	round up (5 5)
●	0.00587 to 2 s.f. = 0.01	counted the zeros as significant
●	3462 to 2 s.f. = 3400	should round up, not down

Key idea

Count significant figures from the first non-zero digit.
Round using normal rounding rules.

0.00473 to 2 s.f. = 0.0047

Leading zeros are placeholders, not significant

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - rounding whole numbers

Round 5738 to 2 significant figures.

First 2 s.f.: 5, 7

Next digit is 3 (< 5), so round down
= 5700

= 5700

Fill remaining places with zeros to keep the number the right size.

Example 2 - rounding decimals

Round 0.04562 to 2 significant figures.

Skip leading zeros: first s.f. is 4,
second is 5

Next digit is 6 (≥ 5), so round up
= 0.046

= 0.046

Leading zeros are not counted as significant.

Example 3 - rounding to 3 s.f.

Round 24,850 to 3 significant figures.

First 3 s.f.: 2, 4, 8

Next digit is 5 (≥ 5), so round up
= 24,900

= 24,900

The 8 becomes 9 because the next digit is 5.

Example 4 - rounding to 1 s.f.

Round 0.000394 to 1 significant figure.

Skip leading zeros: first s.f. is 3
Next digit is 9 (≥ 5), so round up
= 0.0004

= 0.0004

Only one non-zero digit remains in the answer.

YOUR TURN – PRACTICE (deliberate practice)

1. Rounding to 1 s.f.

Round to 1 significant figure.

- a) 74
- b) 386
- c) 4520
- d) 0.067

2. Rounding to 2 s.f.

Round to 2 significant figures.

- a) 473
- b) 6.85
- c) 0.0342
- d) 15800

3. Rounding to 3 s.f.

Round to 3 significant figures.

- a) 4567
- b) 0.05678
- c) 123456
- d) 3.0045

4. Mixed rounding

Round to the number of significant figures stated.

- a) 8765 to 1 s.f.
- b) 0.00456 to 2 s.f.
- c) 345.67 to 3 s.f.
- d) 0.9996 to 3 s.f.

5. Significant figures in context

Round each value appropriately.

- a) A distance of 12,345 m to 2 s.f. b) A mass of 0.00872 kg to 2 s.f. c) A population of 67,530 to 3 s.f. d) A speed of 345.8 m/s to 2 s.f. e) A probability of 0.4567 to 3 s.f.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 0.0047 has 2 significant figures ☐

Correct answer: _____

Reason: _____

b) The zero in 3.02 is not significant ☐

Correct answer: _____

Reason: _____

c) 5738 to 2 s.f. = 5700 ☐

Correct answer: _____

Reason: _____

d) Leading zeros count as significant figures ☐

Correct answer: _____

Reason: _____

e) When rounding up 9, it becomes 10 and carries ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A calculator shows **0.007142857**. Write this to: (i) 1 s.f., (ii) 2 s.f., (iii) 3 s.f. Which is the most useful for a final answer? Explain.

Expression: _____

Simplified: _____

b) The exact area of a circle with radius **4.7 cm** is **69.397...** cm². Round to 3 s.f. If you used this in a further calculation that multiplied by **12**, what would the difference be between using the exact and rounded values?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

